REVIEW

Efficacy and Safety of Perampanel in Patients With Brain Tumor-Related Epilepsy

A Systematic Review and Meta-Analysis

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Abstract

Purpose of Review

Epileptic seizures are a common and often debilitating complication in patients with brain tumors, significantly affecting quality of life and clinical outcomes. Brain tumor–related epilepsy (BTRE) is characterized by complex pathophysiologic mechanisms, notably glutamate-mediated neuronal hyperexcitability within the peritumoral environment. Perampanel, a selective, noncompetitive antagonist of the AMPA glutamate receptor, represents a mechanistically targeted antiseizure medication that may be particularly suited to this context. This review aims to systematically evaluate and synthesize the available evidence regarding the efficacy and safety of perampanel in patients with BTRE, focusing on seizure control, oncologic outcomes, and tolerability in real-world clinical settings.

Recent Findings

A systematic search of PubMed, Embase, and Cochrane Library identified 8 observational studies involving a total of 382 patients with brain tumors and associated seizures treated with perampanel. Across studies, 57.0% of patients were male, and the mean participant age ranged from 43 to 58 years. Pooled analyses demonstrated that 42% (95% CI 27%–59%) of patients achieved seizure freedom, while 46% (95% CI 31%–61%) experienced a ≥50% reduction in seizure frequency. Oncological progression was reported in 39% (95% CI 17%–65%) of patients, with no evidence of perampanel effect on tumor behavior. Adverse events occurred in 26% (95% CI 17%–38%) of patients, most of which were mild to moderate and consistent with the known safety profile of perampanel.

Summary

Perampanel is associated with meaningful seizure control in a substantial proportion of patients with brain tumor–related epilepsy, including high rates of seizure freedom and significant seizure reduction. Importantly, its use does not appear to influence oncologic progression nor substantially increase the incidence of adverse events. Although limited by the nonrandomized nature of the available studies, these findings support perampanel as a valuable therapeutic option for symptomatic seizure management in BTRE, with potential utility in both adjunctive treatment strategies and palliative care settings. Further prospective and controlled studies are warranted to better define its optimal role and long-term impact in this complex patient population.

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Supplementary Material

Introduction

Brain tumor–related epilepsy (BTRE) is a complex neurologic condition that arises as a consequence of structural brain lesions caused by tumors.[1,2] As defined by the International

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League Against Epilepsy (ILAE), tumor-associated epilepsies constitute etiology-specific syndromes that require individualized therapeutic and prognostic consideration.[3] BTRE most commonly presents as focal seizures, which may be aware (motor or nonmotor), impaired awareness, or evolve into bilateral tonic-clonic episodes. Individuals who experience a first unprovoked seizure in the presence of a brain tumor exhibit the highest likelihood of recurrence within the following 2 years.[4]

Perampanel is a first-in-class antiseizure medication (ASM) that functions as a selective, noncompetitive antagonist of postsynaptic α-amino-3-hydroxy-5-methyl-4-isoxazole-propionate (AMPA) receptors.[5,6] By inhibiting AMPA receptor–mediated glutamatergic transmission, perampanel suppresses excitatory neurotransmission implicated in epileptogenesis.[5,6] Electrophysiologic and binding studies demonstrate that perampanel exerts potent inhibitory effects across all AMPA receptor subtypes, while showing negligible activity on N-methyl-D-aspartate (NMDA) and most kainate receptor subunits.[6,7] By contrast, other antiepileptic agents, such as phenobarbital, preferentially inhibit GluA2(R) subunits, and carbamazepine or lamotrigine exhibit NMDA receptor inhibition only at supratherapeutic concentrations.[7] Perampanel did not alter NMDA receptor–mediated currents, and its blockade of nondesensitizing kainate-evoked currents was limited, indicating a high selectivity for AMPA receptor antagonism as its principal mechanism of action.[5] Approved for adjunctive use in patients aged 12 years and older with drug-resistant focal-onset seizures, perampanel has demonstrated broad antiseizure potential and is increasingly being explored in tumor-related epileptic syndromes and generally well tolerated over the longer term in extension studies.[8,9]

Seizures associated with brain tumors are often resistant to standard antiseizure therapies, posing significant challenges for patient management and quality of life.[1,3] In this setting, perampanel provides a distinct pharmacologic mechanism that may contribute to improved seizure control and favorable tolerability, with minimal interaction with antineoplastic treatments.[8,10] This systematic review and meta-analysis seeks to clarify the therapeutic value of perampanel in BTRE by examining current evidence on its efficacy, safety, and clinical outcomes, aiming to delineate its role in the comprehensive management of this patient population.

Methods

This systematic review was conducted in accordance with the methodological guidance of the Cochrane Handbook for Systematic Reviews of Interventions and reported in compliance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement.[11,12] The protocol was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD420251113758).

Eligibility Criteria

Inclusion in this systematic review was restricted to studies that met all of the following criteria: (1) prospective or retrospective observational studies published in English; (2) included adult patients diagnosed with BTRE, defined as epileptic seizures occurring in association with a primary or secondary brain tumor[13,14]; (3) evaluated perampanel, either as monotherapy or adjunctive ASMs therapy; (4) presented single-arm data (no control group); and (5) reported at least one efficacy or safety outcome, including seizure freedom, ≥50% reduction in seizure frequency, oncologic progression, or adverse events (AEs). Studies were excluded if they were nonoriginal (reviews, editorials, commentaries, conference abstracts, theses), animal or preclinical studies, or publications with overlapping populations. In cases of multiple reports from the same registry or cohort, the study with the most complete and updated dataset was retained.

Search Strategy, Study Selection, and Data Extraction

An extensive literature search was conducted in PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) from database inception to April 2025. Full search strategies are available in eMethods 1. Reference lists of all included articles and relevant reviews were manually screened for additional eligible studies. Extracted variables included study design, country, publication year, sample size, tumor type and location, molecular markers (IDH1, MGMT status), seizure characteristics, perampanel dosage, treatment duration, concomitant ASMs, and follow-up period. Two authors (A.M.P.S. & O.R.G.) independently performed title and abstract screening, full-text review, and data extraction. Any disagreement was resolved through discussion with a third author (G.S.N.).

Endpoints Definition and Outcomes

The efficacy outcomes evaluated in this meta-analysis were: (1) Seizure freedom, defined as the proportion of patients who remained seizure-free during the follow-up period, as reported by each study; (2) ≥ 50% reduction in seizure frequency, defined as the proportion of patients achieving at least a 50% reduction in baseline seizure frequency; and (3) oncologic progression, defined as radiologically or clinically confirmed tumor progression during perampanel therapy. The safety and tolerability outcomes included: (1) AEs, defined as any newly occurring or worsening clinical symptom after perampanel initiation. Meta-regression analyses were conducted for all outcomes to evaluate the relationship between the following variables: (1) Study design; (2) sample size; (3) age; and (4) IDH mutation status.

Risk of Bias

Risk of bias was assessed using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Case Series,[15] The checklist evaluates methodological domains related to participant selection, measurement validity, completeness of inclusion, reporting quality, and statistical analysis. Each item was rated as "yes," "no," or "unclear" according to JBI guidance. Methodological details of the JBI approach are provided in eMethods 2. Two authors (A.U.K. & G.M.) independently performed the assessments, with disagreements resolved by consensus or consultation with a third reviewer (A.M.P.S.).

Statistical Analysis

Meta-analyses of single proportions were performed to estimate pooled event rates (%) and corresponding 95% CIs using random-effects models based on the restricted maximum likelihood (REML) estimator. Sensitivity analyses using a generalized linear mixed model (GLMM) were performed to assess the robustness of the pooled estimates (See eTable 2 and eMethods 3). Between-study heterogeneity was quantified using the I^2 statistic and Cochran's Q test, interpreted as low (<40%), moderate (40%–60%), or substantial (≥60%) heterogeneity.[16] Sensitivity analyses were conducted using leave-one-out procedures to assess the influence of each study on pooled estimates and heterogeneity metrics.[16] Publication bias was assessed using funnel plots and the Egger test.[17] Meta-regression analyses were performed on logit-transformed proportions using random-effects models with REML estimation. The following moderators were examined: study design, sample size, mean age, and IDH mutation status. Continuous variables were entered as study-level covariates. Statistical significance was set at a two-tailed $p < 0.05$.[16] All analyses were performed in R version 4.2.2 (R Foundation for Statistical Computing, Vienna, Austria).[18]

Results

Study Selection

A total of 1,602 records were identified from PubMed (n = 234), Embase (n = 1,349), and Cochrane Library (n = 19). After removal of 438 duplicates, 1,164 records underwent title and abstract screening. Of these, 13 reports were retrieved for full-text assessment. Following full-text evaluation, 5 reports were excluded for the following reasons: conference abstract only (n = 5). Ultimately, 8 studies[19-26] met the eligibility criteria and were included in the systematic review and meta-analysis. This study selection process is presented in Figure 1.

Baseline Characteristics of the Included Studies and Patients

Eight observational studies published between 2017 and 2024 were included, comprising a total of 382 subjects with BTRE. Overall, 218 subjects (57.0%) were male, and the mean age across studies ranged from 43 to 58 years. Regarding tumor histology, the prevalence of LGG ranged from 17% to 50%,[22,25] HGG from 25% to 80%,[21,25] and GBM from 16.7% to 27.2%.[24,25] Tumor localization was most frequently frontal (25%–54.5%),[20,24] followed by temporal (11.5%–28.3%)[23,26] and parietal (8.3%–25%)[19,20] sites. Molecular profiling, when reported, indicated IDH1 mutations in 16.7%–62.5%[19,20] of cases and nonmutated IDH1 in 25%–63.6%.[20,24] The average seizure frequency before treatment varied between 9.1 and 13.6 episodes per month.[19,24] The main study and participant characteristics are summarized in Table 1.

Risk of Bias of Included Studies

Risk of bias was assessed using the JBI Critical Appraisal Checklist for Case Series[15] (eTable 2). Most studies fulfilled most methodological domains, including clear inclusion criteria and valid identification of the condition.[19,22-25] Concerns were primarily related to incomplete or nonconsecutive inclusion of participants, particularly in retrospective series,[20,21,24] and to limited reporting of site or clinic demographic information.[19,21,22,25] Statistical analysis was appropriate in prospective cohorts,[19,23] whereas it was limited or insufficiently detailed in smaller retrospective studies.[20-22,24,25] Visual inspection of funnel plots did not suggest publication bias for any outcome; however, interpretation is limited given the small number of included studies (k < 10)[11] (eFigures 1–12).

End Points and Outcomes

Efficacy Outcomes

Data from 7 studies[19,20,22-26] were included for seizure freedom at follow-up, with a pooled rate of 42% (95% CI 27%–59%; I^2 = 65.7%; Figure 2A) under the random-effects model. Between-study heterogeneity was substantial. Eight studies[19-26] provided data on ≥ 50% reduction in seizure frequency, presenting a pooled rate of 46% (95% CI 31%–62%; I^2 = 84.3%; Figure 2B), indicating substantial heterogeneity. Five studies[19,21-24] reported data on oncologic progression during perampanel therapy; the pooled incidence was 39% (95% CI 17%–65%; I^2 = 0.0%; Figure 3), suggesting stability across the included samples.

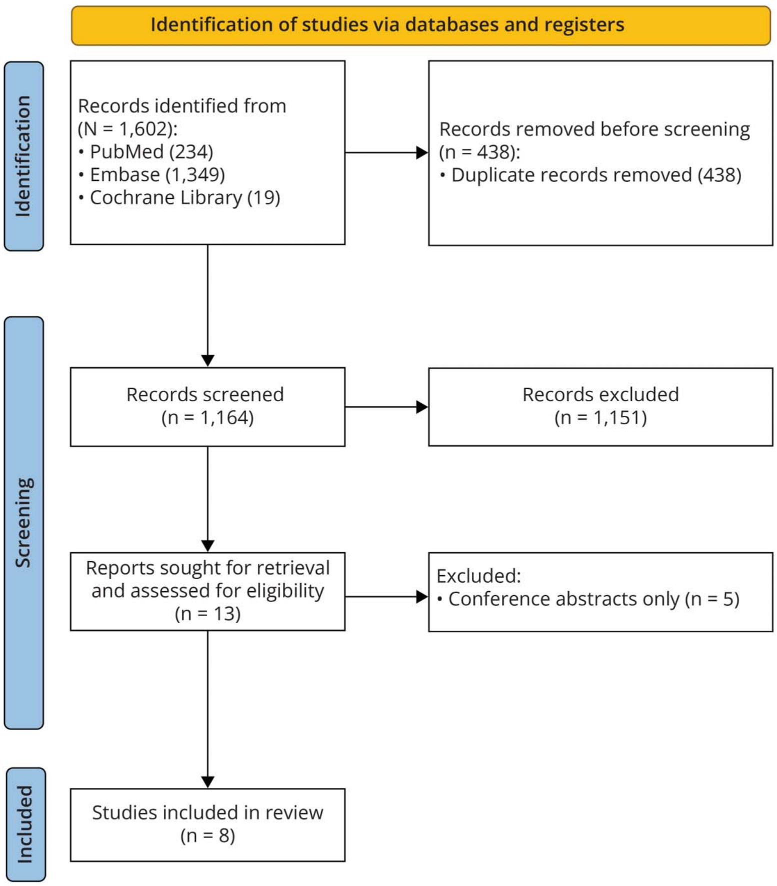
Safety Outcomes

A total of 8 studies[19-26] reported data on any AEs, with a pooled incidence of 26% (95% CI 17%–38%; I^2 = 61.6%; Figure 4), indicating moderate heterogeneity. The most frequently reported AEs were dizziness, somnolence, and fatigue, consistent with the known safety profile of perampanel.

Sensitivity Analysis, Publication Bias, and Meta-Regression

Leave-one-out sensitivity analyses confirmed the robustness of all outcomes. For seizure freedom, pooled proportions ranged from 0.37 to 0.50; omitting Coppola et al.[19] resolved the heterogeneity (I^2 = 0%) and had the largest influence on overall heterogeneity. For ≥50% seizure reduction, the results were stable (0.38–0.49) and showed no heterogeneity (I^2 = 0%) when excluding Nakai et al.[26]; the latter study exerted the strongest influence. For oncologic progression, pooled proportions ranged from 0.32 to 0.44 with consistently low heterogeneity (I^2 = 0); Izumoto et al.[22] had the

Figure 1 Prisma Flow Diagram of Study Screening and Selection



largest impact on the overall effect estimate. For adverse events, the results remained stable (0.26–0.30) with moderate heterogeneity (I^2 = 36.3–67.0%); Nakai et al.[26] contributed the most to the overall results. Sensitivity analyses using GLMM produced comparable pooled estimates across all outcomes (eTable 1). Study design was the only significant moderator for seizure freedom rates (coefficient = 1.03; p = 0.0345), with retrospective studies reporting higher values. No variable influenced ≥50% seizure reduction, and none explained heterogeneity in oncologic progression. For AEs, study design (coefficient = −1.03; p = 0.0398) and IDH mutation status (coefficient = −1.57; p = 0.0441) were significant, indicating lower event rates in retrospective cohorts and higher probabilities in IDH-mutant cases. Sample size and mean age showed no association with any of the outcomes. Full estimates are presented in Table 2 and eFigures 13–27.

Discussion

BTRE represents a clinically challenging interface between neuro-oncology and epilepsy, in which seizure control is frequently refractory and must be balanced against oncologic treatment constraints. Despite the increasing use of newer non–enzyme-inducing ASMs, evidence guiding optimal pharmacologic strategies in this population remains limited and predominantly observational. Within this context, clarifying the real-world efficacy and safety profile of mechanistically targeted agents such as perampanel is of direct clinical relevance.

In this systematic review and meta-analysis of 8 observational studies,[19-26] perampanel use in patients with BTRE was associated with clinically meaningful seizure control and an acceptable safety profile. Approximately one-third of patients achieved seizure freedom, while a similar proportion experienced a ≥50% reduction in seizure frequency. Oncological progression occurred in a substantial proportion of patients, with highly consistent estimates across cohorts, and AEs were generally mild to moderate, aligning with the established tolerability profile of perampanel.

The observed seizure freedom rate indicates that complete seizure control can be achieved in a clinically relevant subset of patients with BTRE treated with perampanel, despite the intrinsic pharmacoresistance characteristic of tumor-related epilepsies. The pooled estimate for ≥50% seizure reduction

Table 1 Baseline Characteristics of Included Studies

Authors, y	Study design	Sample size, n (%)	Age, y Mean ± SD or mean	Male, n (%)	Tumor types, n (%)	Tumor site, n (%)	IDH1 mutation status, n (%)	MGMT methylation status, n (%)	Seizure types, n (%)	Seizure frequency, mean ± SD per month	Monotherapy/ Polytherapy (%)	Mean per dose (mg/d)
Izumoto et al. (2018)[22]	Retrospective, Observational study	12 (100)	58.0 ± 39.3	8 (67.0)	LGG: 2 (17.0); HGG: 8 (67.0); GBM: 2 (17.0)	NA	NA	NA	Partial: 8 (67); Generalized: 4 (33)	NA	NA	5.2
Maschio et al. (2018)[24]	Retrospective, Observational study	11 (100)	54	9 (82.0)	HGG: 4 (36.4); LGG: 4 (36.4); GBM: 3 (27.2)	Frontal: 6 (54.5); Temporal: 3 (27.3); Other: 2 (18.2)	Mutated: 3 (27.3) Not mutated: 7 (63.6); Unknown: 1 (9.1)	Methylated: 5 (45.5) Unmethylated: 5 (45.5); Unknown: 1 (9)	Partial: With second generalization 6 (54.5); Generalized: 5 (45.5)	13.6 ± 18.7	Mono: 45.5; Poly: 54.5	7.45
Vecht et al. (2017)[25]	Prospective, Observational Study	12 (100)	45.6 ± 25.2	9 (75.0)	LGG: 6 (50.0); HGG: 3 (25.0); GBM: 2 (16.7); Ganglioglioma: 1 (8.3)	NA	NA	NA	Partial: 7 (58); Complex partial: 4 (33); Generalized: 1 (8.3); Partial status epilepticus: 2 (16.7)	NA	Mono: 0; Poly: 100	7.17
Coppola et al. (2020)[19]	Prospective, Observational Study	36 (100)	45.3 ± 44.47	23 (63.9)	LGG: 11 (30.6); HGG: 14 (38.9); GBM: 7 (19.4); Other: 4 (11.1)	Frontal: 12 (33.3); Temporal: 6 (16.7); Parietal: 3 (8.3); Multilobular: 14 (38.9); Occipital: 1 (2.8)	Mutated: 6 (16.7); Not mutated: 10 (27.8); Unknown: 20 (55.6)	Methylated: 7 (19.4); Unmethylated: 4 (11.1); Unknown: 25 (69.5)	Focal with preserved awareness: 14 (38.9); Focal with compromised awareness 7 (19.4); Focal to bilateral: 11 (30.6); Generalized: 4 (11.1)	9.1 ± 12.8	Mono: 52.8; Poly: 47.2	6.53
Heugenhauser et al. (2021)[21]	Retrospective, Observational Study	5 (100)	43.0 ± 15.7	3 (60.0)	HGG: 4 (80); Astrocytoma: 1 (20)	Temporal: 1 (20); Central-parietal: 1 (20); Fronto-parietal: 2 (40); Frontal multilobular: 2 (20)	Not mutated 3 (60)	Methylated: 1 (20); Unmethylated: 1 (20)	Focal to bilateral tonic-clonic: 4 (80); Focal aware and impaired awareness: 1 (20)	NA	Mono: 0; Poly: 100	6.8
Maschio et al. (2020)[23]	Retrospective, Observational Pilot Study	26 (100)	47.5	16 (61.5)	LGG: 8 (30.77); HGG: 8 (30.77); GBM: 7 (27); Meningiomas: 2 (7.7); Metastasis: 1 (4.0)	Frontal: 8 (30.7); Multilobular: 8 (30.7); Temporal: 3 (11.5); Parietal 6 (23.1); Occipital: 1 (4)	Mutated: 6 (23.1); Not Mutated: 9 (34.6); Unknown: 11 (42.3)	Methylated: 7 (26.9); Unmethylated: 4 (15.4); Unknown: 15 (57.7)	Focal aware: 13 (50); Focal to bilateral tonic-clonic: 7 (27); Focal unaware: 6 (23)	10.6 ± 14.27	Mono: 42.3; Poly: 57.7	6.4
Dunn-Pirio et al. (2018)[20]	Prospective, Single-Arm, Open-Label Study	8 (100)	45.12	6 (75.0)	HGG: 4 (50); LGG: 4 (50)	Frontal: 2(25); Frontotemporal:1 (12.5); Parietal: 2 (25); Temporal: 1 (12.5); Frontoparietal: 2 (25)	Mutated: 5 (62.5); Not mutated: 2 (25); Unknown: 1 (12.5)	Methylated: 7 (87.5); Unknown: 1 (12.5)	Focal impaired awareness: 8 (100)	NA	Mono: 25; Poly: 75	NA

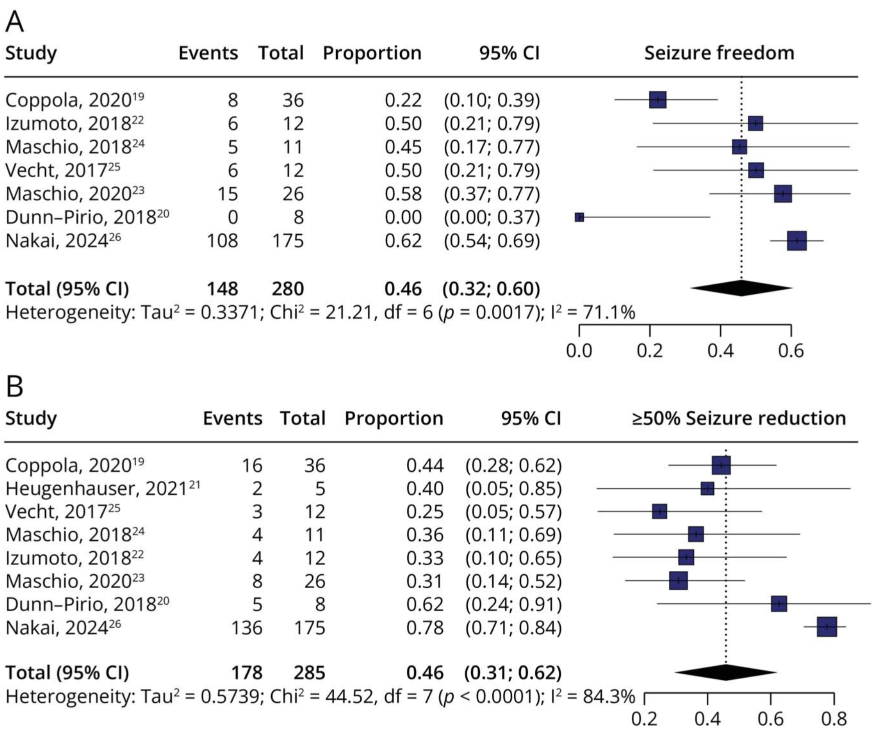
Continued

Table 1 Baseline Characteristics of Included Studies *(continued)*

Authors, y	Study design	Sample size, n (%)	Age, y Mean ± SD or mean	Male, n (%)	Tumor types, n (%)	Tumor site, n (%)	IDH1 mutation status, n (%)	MGMT methylation status, n (%)	Seizure types, n (%)	Seizure frequency, mean ± SD per month	Monotherapy/ Polytherapy (%)	Mean per dose (mg/d)
Nakai et al. (2024)[26]	Prospective, Observational Study	272 (100)	NA	144 (52.9)	NA	Frontal: (41.2); Temporal: (28.3) Parietal: (19.5); Occipital: (4.0); Multilobed: (4.8); Unknown: (2.2)	NA	NA	Focal to bilateral tonic–clonic seizure: (20.2); Focal impaired awareness seizure: (34.2); Focal aware seizure: (29.8); Unknown: (15.8)	NA	Mono: 57.0; Poly: 43.0	3.36

Abbreviations: GBM = glioblastoma; HGG = high-grade-glioma; IDH1 = Isocitrate dehydrogenase 1; LGG = low-grade-glioma; MGMT = O6-Methylguanine-DNA-methyltransferase; n = number; NA = not available; PER = perampanel; PNET = primitive neuroectodermal tumor.

Figure 2 Pooled Rates of (A) Seizure Freedom, and (B) ≥50% Reduction in Seizure Frequency



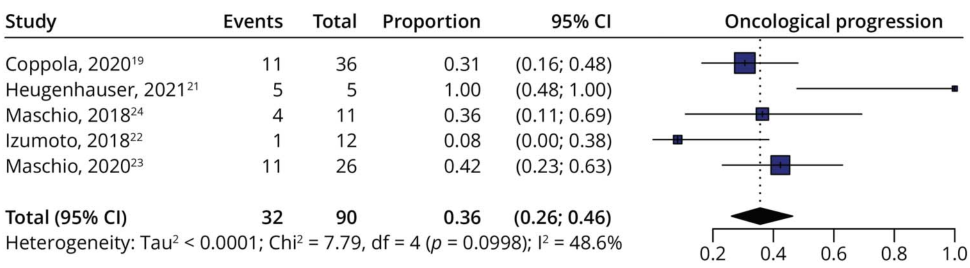
Squares represent the weight of each study, with larger squares indicating studies with greater weight and precision. The black diamond indicates the overall pooled effect derived from the included studies.

further supports its clinical utility, particularly as this end point reflects a meaningful reduction in seizure burden and potential improvement in quality of life in refractory populations. When contextualized within the existing literature, these findings are consistent with prior systematic reviews reporting favorable seizure outcomes with perampanel, while providing more conservative and generalizable estimates derived from heterogeneous, real-world cohorts.[10,27] Comparatively, response rates are broadly in line with those reported for other contemporary non–enzyme-inducing agents, such as levetiracetam and lacosamide, in glioma-associated epilepsy, positioning perampanel as a competitive therapeutic option within current clinical practice.[28,29]

Interpretation of efficacy outcomes must take into account the observed heterogeneity patterns. Although seizure freedom showed high heterogeneity, it was nullified upon removal of Coppola et al.[19] in the sensitivity analysis, suggesting sensitivity to study-level characteristics such as design, patient selection, baseline seizure burden, and oncologic control. A similar trend was observed regarding ≥50% seizure reduction, indicating a highly consistent treatment effect across diverse clinical settings.

Meta-regression analyses identified study design as a significant moderator of seizure freedom, with retrospective studies reporting higher rates than prospective cohorts. Given the descriptive nature of the included studies, this difference likely reflects variations in reporting practices, patient selection, and follow-up assessment rather than true differences in treatment effect. For AEs, both study design and IDH mutation status were associated with variability in

Figure 3 Pooled Rates of Oncological Progression

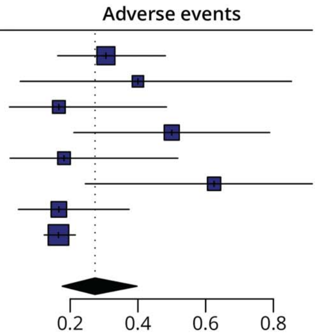


Squares represent the weight of each study, with larger squares indicating studies with greater weight and precision. The black diamond indicates the overall pooled effect derived from the included studies.

Figure 4 Pooled Rates of Adverse Events

Study	Events	Total	Proportion	95% CI
Coppola, 2020[19]	11	36	0.31	[0.16; 0.48]
Heugenhauser, 2021[21]	2	5	0.40	[0.05; 0.85]
Izumoto, 2018[22]	2	12	0.17	[0.02; 0.48]
Vecht, 2017[25]	6	12	0.50	[0.21; 0.79]
Maschio, 2018[24]	2	11	0.18	[0.02; 0.52]
Dunn−Pirio, 2018[20]	5	8	0.62	[0.24; 0.91]
Maschio, 2020[23]	4	24	0.17	[0.05; 0.37]
Nakai, 2024[26]	45	272	0.17	[0.12; 0.22]
Total (95% CI)	**77**	**380**	**0.27**	**[0.18; 0.40]**

Heterogeneity: $Tau^2 = 0.3332$; $Chi^2 = 18.23$, df = 7 ($p = 0.0110$); $I^2 = 61.6\%$



Squares represent the weight of each study, with larger squares indicating studies with greater weight and precision. The black diamond indicates the overall pooled effect derived from the included studies.

reported rates; however, these analyses were exploratory, based on aggregate data, and should be interpreted cautiously. No covariate explained variability in ≥50% seizure reduction or oncologic progression, reinforcing the robustness and consistency of these outcomes across studies. Sensitivity analyses confirmed the stability of all pooled estimates, with leave-one-out analyses demonstrating that no single study materially influenced the overall results. Although individual studies contributed variably to pooled proportions, the direction and magnitude of effects remained consistent. Visual inspection of funnel plots[17] did not suggest overt publication bias[11]; however, the limited number ($K < 10$) of included studies substantially restricts the reliability of such assessments and warrants cautious interpretation.

The observed clinical effects are biologically plausible given perampanel's selective antagonism of AMPA receptors, a central driver of glutamate-mediated hyperexcitability in tumor-related epileptogenesis.[9,30] Excessive glutamatergic signaling, together with peritumoral hypoxia, acidosis, and blood-brain barrier disruption, contributes to seizure generation and persistence in BTRE.[30-32] While AMPA receptor pathways have been implicated in experimental glioma biology, the current clinical data support a role for perampanel in seizure control rather than tumor modification.[33,34]

Table 2 Meta-Regression Summary of Results

Outcome	Predictor	Coefficient (Coef)	SE	z	*p* Value
Adverse events	Study design	−1.0274	0.4998	−2.056	0.0398
	Sample size	−0.6049	0.6111	−0.99	0.3223
	Mean age	−0.9697	0.7087	−1.368	0.1712
	IDH mutation status	−1.567	0.7783	−2.013	0.0441
≥50% seizure reduction	Study design	−0.4216	0.4006	−1.052	0.2926
	Sample size	0.0517	0.4052	0.128	0.8985
	Mean age	−0.1988	0.4924	−0.404	0.6864
	IDH mutation status	−0.9723	0.767	−1.268	0.205
Seizure freedom	Study design	1.0316	0.488	2.114	0.0345
	Sample size	−0.1516	0.7361	−0.206	0.8368
	Mean age	0.4411	0.7646	0.577	0.564
	IDH mutation status	2.4434	1.7011	1.436	0.1509
Oncological progression	Study design	0.3708	1.2203	0.304	0.7612
	Sample size	0.0002	0.9458	0	0.9999
	Mean age	−0.9881	0.9461	−1.044	0.2963

Abbreviations: IDH1 = isocitrate dehydrogenase 1; SE = standard error.

From a clinical perspective, these findings support consideration of perampanel as a nonenzyme-inducing ASM for patients with BTRE, particularly in settings where drug-drug interactions with oncologic therapies must be minimized.[35,36] The consistency of seizure reduction outcomes suggests that perampanel may be especially valuable as adjunctive therapy aimed at reducing seizure burden rather than guaranteeing complete seizure freedom. From a research standpoint, the exploratory signals identified through meta-regression highlight the need for prospective studies incorporating standardized outcome definitions, molecular tumor characterization, and patient-level predictors of response.

Limitations

This meta-analysis is subject to limitations inherent to the available evidence. All included studies were single-arm observational series, precluding causal inference and preventing assessment of comparative efficacy. The absence of control groups limits evaluation of confounding and does not allow estimation of relative treatment effects. According to the JBI critical appraisal,[15] several studies did not clearly document consecutive or complete inclusion of participants, raising the possibility of selection bias. Heterogeneity in outcome definitions, duration of follow-up, and reporting of AEs may have introduced measurement and reporting bias. Sample sizes were small, follow-up periods were relatively short, and reporting of perampanel dose, treatment duration, and concomitant therapies was inconsistent across studies. Restriction to English-language publications may have introduced language bias. In addition, CIs for some pooled estimates were wide, reflecting limited sample sizes. Finally, the modest number of studies limited the statistical power of meta-regression analyses and the assessment of publication bias. These limitations should be considered when interpreting subgroup analyses and exploratory findings.

Conclusion

In summary, this meta-analysis indicates that perampanel provides clinically meaningful seizure control in a substantial proportion of patients with BTRE, with consistent seizure reduction and an acceptable safety profile. The observed rate of oncologic progression is consistent with the expected course of the underlying tumors and does not suggest a disease-modifying oncologic effect. Perampanel should therefore be regarded as an effective symptomatic and palliative option within multimodal neuro-oncologic care. Further prospective studies with longer follow-up and integrated molecular characterization are needed to refine patient selection and optimize therapeutic strategies in this complex population.

Author Contributions

A.M.P. Da Silva: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. O.R. Gonçalves: study concept or design; analysis or interpretation of data. G.C.N. Tudella: major role in the acquisition of data; study concept or design; analysis or interpretation of data. G. Menegucci: study concept or design; analysis or interpretation of data. A.U. Klostermann: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M.E.S. Valverde: major role in the acquisition of data; study concept or design. P.F. Meldola: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. A. Gunkan: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. G.S. Noleto: drafting/revision of the manuscript for content, including medical writing for content; study concept or design.

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